22.2. RADIONUCLIDES OF ENVIRONMENTAL CONCERN

Most of the radionuclides produced in nuclear tests, accidents and in the normal fuel cycle are short lived. In Table 21.3, longer lived fission products and activation products from these systems are listed as these represent the major concern to the general public if they are allowed to enter the environment; we exclude nuclides with insignificant contribution to the total activity after 10 years. In addition, several heavy radionuclides are formed by neutron capture reactions such as ^{236}U, ^{237}Np, $^{238-242}$Pu, ^{241}Am, etc.

To assess the potential for these radionuclides to cause harm to humans, their geochemical and biological behavior must be evaluated. For example, since Kr is a chemically inert gaseous element, it would have little effect on a person who inhaled and immediately exhaled the small amount which might be present in the air. By contrast, other nuclides with high activity, ^{90}Sr–^{90}Y and ^{137}Cs–^{137m}Ba, have active geological and biological behavior and can present much more significant radiation concerns to humans. In normal operations, these nuclides would of course be released in quite insignificant amounts, as represented by the value for aquatic releases in reactor operation and reprocessing, Table 19.7 and 21.10. Similarly, the heavy elements (Np, Pu, Am) would not be released in normal operation at levels that would be of concern. We have pointed out that some low level releases as a result of normal operations are allowed by the health authorities, who also monitor these levels. In the effluents from reprocessing plants (e.g. Sellafield in the United Kingdom and La Hague in France), the relatively long-lived nuclides such as ^{3}H, ^{14}C, ^{85}Kr, ^{99}Tc and ^{129}I are of major concern. The liquid effluents from nuclear power plants and from reprocessing plants are about equally responsible for the global collective dose commitment of nuclear power generation (i.e., $\sim$0.8 man Sv per GW_ey of the total $\sim$2.5 man Sv).

22.3. RELEASES FROM LARGE REACTOR ACCIDENTS

So far there have only been two major nuclear reactor accidents, Chernobyl and Fukushima. In both cases large amounts of radioactive material was released into the environment. However, in these events there was only an insignificant addition of ^{14}C to the atmosphere relative to the average atmospheric amount of ^{14}C at the corresponding times.

22.3.1. Chernobyl

On April 26, 1986, a low power engineering test was being conducted at one of the reactors of the Chernobyl nuclear power station in the Ukraine (then the USSR). The reactor became unstable, resulting in thermal explosions and fires that caused